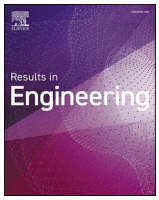
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Case report



Photovoltaic panel cooling using ground source energy: CFD simulation

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ABSTRACT

The performance of solar panels is significantly affected by high temperatures, leading to various cooling methods being employed to enhance their efficiency. This study utilizes comprehensive computational fluid dynamics (CFD) simulations with Ansys Fluent 2023 R1 software to evaluate the effectiveness of cooling photovoltaic (PV) panels using ground source energy in Kirkuk city, northern Iraq (35° 28′ N - E. 44.39). A 3-D model was employed to analyze the thermal behavior of the PV panel over 10 h (8:00 a.m. to 5:00 p.m.). The simulation demonstrates that the proposed cooling method achieved an impressive temperature reduction of approximately 28 % compared to conventional cooling methods. Moreover, the PV panel with ground-source cooling consistently outperformed the PV-only setup, showing a noteworthy improvement of approximately 6.5 % in power output, reaching a maximum of around 88 W. Examination of velocity and temperature distribution contours revealed that ambient air drawn in as hot air gradually dissipated its heat to the surrounding soil and bricks as it traversed through underground pipes, resulting in a significant reduction in temperature. This cooling process effectively enhanced the overall performance of the PV panel.

1. 1-Introduction

The problem of continuous energy availability is one of the most important problems facing humanity currently. Recent statistics indicate the possibility of the depletion of traditional fuels within the next fifty years [1]. Scientists and engineers are trying to search for new sources of energy that are not polluting the environment and are not depleted in the near future [2]. Now, renewable energies include solar energy, wind energy, tidal energy, water dam energy, etc [3]. However, solar energy is the most important source of renewable energy that can replace oil due to its availability in most regions of the world and its lack of advanced technology for use [4]. There are currently many solar energy systems available, such as thermal solar collectors, solar chimneys [5], PV/Trombe walls [6], solar ponds [7], and solar distillers [8], but PV panels have taken the lead in generating electricity using solar energy [9]. The rapid growth and adoption of photovoltaic (PV) technology for solar power generation have positioned it as a key player in the global transition towards clean and sustainable energy sources [10]. As PV panels convert sunlight into electricity, they are subjected to various environmental factors that can impact their performance, including temperature. The temperature of PV panels has a direct influence on their efficiency and lifespan, making effective cooling strategies crucial for optimizing their output and ensuring long-term reliability [9]. PV panels are known to exhibit a decrease in efficiency as their temperature rises, following a characteristic temperature coefficient [11]. As the panels heat up, the semiconductors within them experience an increase in electron mobility, leading to a drop in the voltage output. Consequently, the overall energy conversion efficiency of the PV system is compromised [12]. Additionally, prolonged exposure to high temperatures can accelerate the degradation of the panel's materials, reducing their operational lifespan and overall performance [13]. To mitigate these challenges, cooling techniques are employed to maintain PV panels within an optimal temperature range [14].

Various cooling methods have been explored, including passive cooling, active cooling, and hybrid cooling systems. Passive cooling relies on natural heat dissipation mechanisms such as convection, radiation, and conduction to remove excess heat from the panels. Active cooling utilizes mechanical systems, such as fans or liquid-based cooling, to actively circulate a cooling medium and enhance heat dissipation. Hybrid cooling systems combine both passive and active techniques to maximize cooling efficiency. Passive cooling methods are often cost-effective and simple to implement since they do not require

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additional energy inputs [15]. To increase the amount of electricity that PV panels are capable of producing, the researchers in this study made use of a variety of cooling strategies. Bjørk and Nielsen [16] evaluated the performance of several commercial PVs and TEG, whereas Soltani et al. [17] experimentally explored five different techniques for a hybrid system with nanofluids, and found that SiO_2/water performed the best of the five. Both groups concluded that SiO_2/water was the most effective.

Zhang and Xuan [18] conducted an experimental investigation in which they investigated various configurations of the PV-TE system. These configurations included PV-TE by itself, PV-TE without ceramic plates, and PV-TE with a V-type groove. The contradictory behavior of thermoelectric load resistance in photovoltaic-thermoelectric modules was investigated by Li et al. [19], who concluded that the load resistance of TE alone generates the highest possible amount of power production. A numerical analysis was carried out by Kant et al. [20] which highlighted the significant impact that the inclination angle and heat transport in PCM have on the operating temperature of PV panels. An experimental investigation on a PV-PCM system was conducted by Hasan et al. [21] utilizing the Simulink Toolbox of MATLAB. The researchers found that the PV yearly energy efficiency increased by 5.9 % under hot climatic circumstances as a result of their work. Radziemska and Kucharek [22] carried out an experimental investigation making use of PCMs. They discovered that a PV/PCM system operating with a steel tank containing 42–44 paraffin produced the most favorable results. According to the findings of Khanna et al. [23] investigation on the impacts of the operating conditions of the PCM container, the azimuth angle and the container depth have a substantial impact on the amount of energy that can be stored. Nizetic et al. [24] investigated the efficiency, power, and temperature connections of PV panels using various cooling approaches. They found that liquids had the highest efficiency gains of 22 %, while PCM and air both had increases of 21.2 % and air had an increase of 20 %. According to the findings of Machniewicz et al. [25], who investigated how the performance of PV panels changed when PCM was used at one of four distinct melting temperatures, the panels performed at their best when the temperature ranged from 18 to 25 °C. Abdelrahman et al. [26] evaluated the effects of using PCM (RT35HC) coupled with nanoparticles (Al_2O_3), they observed reduced surface temperatures of PV panels whether employing PCM or nanoparticle-reinforced PCM. Atkin and Farid [25] developed four different systems, which they referred to as PV, PV + PCM, PV + fin, and PV + PCM + fin. Based on the findings of both experimental and simulation investigations, Case D produced the most favorable outcomes. Emad J Mahdi et al. [27] evaluated the accuracy of solar ground station data compared to data from the PVGIS, employing empirical equations to develop software for analyzing global, direct, and diffuse solar radiation rates. The analysis assessed data compatibility, dependability, and confidence, revealing variations in solar radiation rates throughout the day at different sites. Despite technical challenges with ground station data, correlations with PV GIS data suggest a potential for reliable estimation in solar energy applications with proper data processing and maintenance of measurement devices. Integration of multiple data sources could enhance station data accuracy. Jafari [28] used a new design of a geothermal heat exchanger to cool the PV panel. According to the experimental findings, the average net electricity generation of the PV panel with a cooling system outperforms the PV panel without a cooling system by 9.8 %. Liu et al. [29] presented multi-criteria evaluations and multi-objective optimizations for a geothermal PV hybrid system assisted by nanofluid. The outcomes demonstrate that spiral pipe and 2 % Al2O3 nanofluid work best together. The monocrystalline solar panels with the heat pump assisted by nanofluids produce the lowest cost, the lowest CO_2 emissions, and the maximum energy efficiency. Lopez-Pascual et al. [30] presented a theoretical study to enhance the performance of the solar cell using a geothermal cooling system. The results confirmed that for July, a peak efficiency improvement of 10.7 % is computed, with an ideal flow rate of 1.8 l/min per square meter of solar module. It also computes a weighted average power gain of up to 4.9 % annually. Mohammed T. Hussain and Emad J. Mahdi [31] conducted a study comparing global solar radiation data with photovoltaic (PV) potential at various sites across Iraq. Utilizing PV GIS data and empirical equations in Visual Basic.NET, the research assessed PV potential maps in Iraq using ArcGIS software. Results indicate many sites are suitable for constructing photovoltaic solar plants due to ample solar radiation. However, site selection must consider geographical and environmental factors, proximity to power transmission lines, urban areas, and workforce availability. Recommendations include constructing photovoltaic solar power plants of at least 1 MW in size based on the PV potential map for Iraq.

After reviewing previously published scientific articles, it is evident that there is a lack of research on utilizing cooled air from the ground to cool PV panels. Most prior studies have concentrated on employing ground-cooled water for this purpose, which may not be feasible in many regions, particularly desert areas. The current study addresses this gap by focusing on using cooled air circulated through underground pipes to cool photovoltaic panels, leveraging Computational Fluid Dynamics (CFD) technology. Additionally, the study aims to compare the effectiveness of this method with other cooling techniques, which are often more costly.

2. 2- Governing equations

In the present analysis, the following assumptions were considered as the basis for the solution: a steady-state method for fluid flow and heat transfer, convection and radiation losses are thermal losses from the PV panel surface, and airflow is turbulent. The continuity, momentum, energy, and standard k-epsilon equations can be written as follows [32]:

Continuity equation:

$$\nabla \cdot (\rho \vec{v}) = 0 \tag{1}$$

Where ρ *and* $\vec{v}$ are the air density and vector velocity, respectively.

Energy equation:

$$\nabla \cdot \left(\vec{v}(\rho E + p) = \nabla \cdot \left(k_{eff} \nabla T + \left(\mu \left[(\nabla \vec{v} + \nabla \vec{v}^T) - \frac{2}{3} \nabla \cdot \vec{v} I \right] \cdot \vec{v} \right) \right) + S_h \right. \tag{2}$$

Where E *and* p are the energy per unit mass and pressure, respectively. k_{eff} is the effective conductivity ($k + k_t$, where k_t is the turbulent thermal conductivity, defined according to the turbulence model being used), μ and S_h are the air viscosity and any volumetric heat source, respectively.

Momentum equation:

$$\nabla \cdot (\rho \vec{v} \vec{v}) = -\nabla p + \nabla \cdot \left(\mu \left[(\nabla \vec{v} + \nabla \vec{v}^T) - \frac{2}{3} \nabla \cdot \vec{v} I \right] \right) + \rho \vec{g} \tag{3}$$

Where T *and* $\vec{g}$ are the temperature and gravitational acceleration, respectively.

K-epsilon equations:

$$\frac{\partial}{\partial x_i}(\rho k u_i) = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\delta_k} \right) \frac{\partial k}{\partial x_j} \right] + G_k + G_b - \rho \varepsilon \tag{4}$$

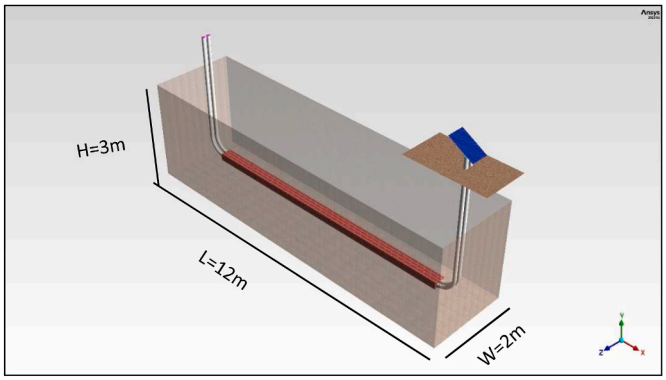
$$\frac{\partial}{\partial x_i}(\rho \varepsilon u_i) = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\delta_\varepsilon} \right) \frac{\partial \varepsilon}{\partial x_j} \right] + C_{1\varepsilon} \frac{\varepsilon}{k} (G_k + C_{3\varepsilon} G_b) - C_{2\varepsilon} \rho \frac{\varepsilon^2}{k} \tag{5}$$

Where $x_i, k, u_i, \mu_t, \delta_k, G_k, G_b, \varepsilon,$ *and* δ_ε are the axial position, turbulent energy, axial velocity, turbulent viscosity, turbulent Prandtl number for k, production of turbulent kinetic energy (TKE) due to mean velocity shear, production of TKE due to buoyancy, turbulent dissipation rate, and turbulent Prandtl number for ε, respectively. constants of the model have the following default values: $C_{1\varepsilon} = 1.44$, $C_{2\varepsilon} = 1.92$, $C_{3\varepsilon} = 0.09$, $\delta_k = 1.0$, $\delta_\varepsilon = 1.3$.

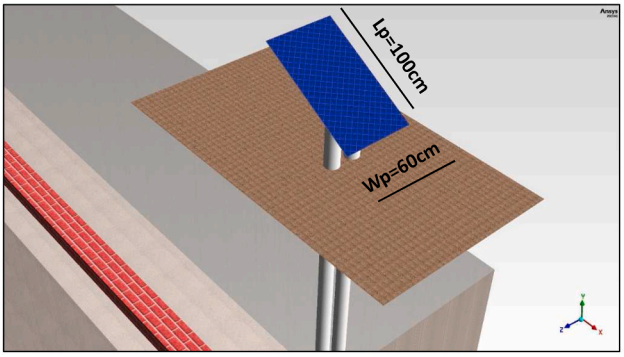
3. 3- Methodology

3.1. 3-1 numerical procedure

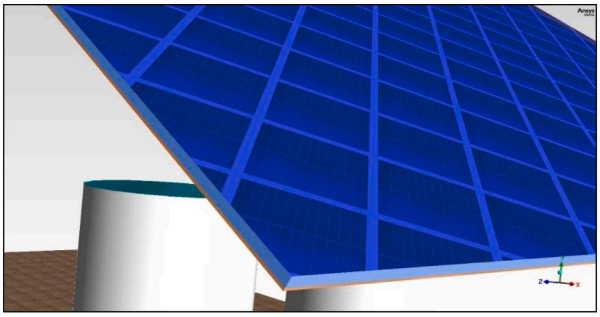
The Ansys Fluent software was utilized to construct a three-dimensional geometry model of the PV panel, incorporating ground source cooling piping and cement. The PV panel measures 100 cm in length and 60 cm in width, with a maximum power output of 100W. The ground area, measuring 12 m in length, 3 m in width, and 3.6 m in height, was modelled as a rectangular area. Two PVC pipes, each with a diameter of 10 cm, were designed, buried underground, and integrated with the ground-source system to enhance heat absorption, as depicted in Figure (1-a). The PV panel comprises six layers: glass covering, Ethylene Vinyl Acetate (EVA) layers 1 and 2, monocrystalline silicon PV cells, Tedlar, and aluminum. In the PV-only system, only the PV panel was modelled and exposed to solar radiation. In contrast, for the air-cooling system, an underground area with bricks and fluid-carrying pipes was constructed and integrated with the PV panel, as shown in Figure (1-b). Figure (1-c) provides a close-up illustration of the airflow process behind the solar panel for further clarification. Table 1 details the material properties of each layer within the PV panel and the ground



(a) Complete designed system showing all the components



(b) Close-look view of the PV geometry with its air domain



(C) Close-look view of the PV layers and pipes connected to it

Fig. 1. Mesh generation of the system.

Table (1)
Materials properties and sizes of each layer [20].

No	Material	Thickness (cm)	Thermal conductivity	Specific Heat Capacity	Density
1	Glass	0.3	1.8	500	3000
2	EVA layers (1,2)	0.05	0.35	2090	960
3	PV cell	0.04	148	677	2330
4	Tedlar	0.01	0.2	1250	1200
5	PVC	0.1	0.15	–	–
6	Soil	300	0.25	–	–
7	Bricks	40x25	0.5	–	–

source system geometry model. Meshing, depicted in Figure (2 – A, B, C), was performed according to the model using ANSYS Meshing software. A grid independence meshing test (GIT) was conducted to ensure the appropriate cell count for achieving mesh-independent results [11].

To calculate the solar panel efficiency, the following equation will be utilized:

$$\eta_p = \eta_r\left[1 - \beta_p(T_{PV} - T_r)\right] \quad (6)$$

Where:

η_p, solar panel efficiency, η_r, solar panel reference efficiency, which is taken as 0.17, β_p, temperature coefficient for the model efficiency (% °C), T_{PV} is the temperature of the solar panel (°C), and T_r is the module reference temperature at 25 °C.

After obtaining the efficiency of the solar panel, we can now calculate the electrical power produced as follows:

$$P_p = \eta_p \times A_p \times G \quad (7)$$

Where:

P_p, solar panel output power (W), A_p, Area of the solar panel (m^2), G, solar radiation (W/m^2).

3.2. 3-2 boundary conditions

Table 2 presents the comprehensive boundary conditions employed in the system under investigation. The inlet of the pipes is defined as the air velocity originating from the ambient environment, while the outlet represents the atmospheric pressure surrounding the PV panel. Additionally, the outer surfaces of the ground are set to maintain a constant temperature. The research objective is to identify a straightforward and economically viable method for cooling solar panels that minimizes energy consumption.

4. 4- Results and discussion

The simulation was conducted using Ansys Fluent 2023 R1 software to create and simulate a three-dimensional model aimed at examining the cooling of PV panels through underground cooled air in Kirkuk City, northern Iraq. The cooling process involves dissipating heat from the panels to the surrounding ground soil and bricks, facilitating increased heat absorption, as illustrated in figures (3-A, B). Consequently, the air is cooled to a sufficient degree to effectively lower the temperature of the PV panels and enhance their performance.

Figure (4) displays the graph representing the Grid Independence Test (GIT), determining the optimal number of cells, which was approximately 3,628,059 cells. The model setup included enabling equations for energy, continuity, momentum, radiation, and turbulence, which were subsequently solved. The radiation aspect was addressed by activating the solar irradiation model utilizing the discrete ordinate model (DO), specifically tailored to Iraqi Kirkuk city, considering its longitude (44.38) and latitude (35.46).

Fig. 5 displays the solar radiation values observed in Kirkuk from 8:00 a.m. to 5:00 p.m. The peak solar radiation, 950 W/m^2, occurred at

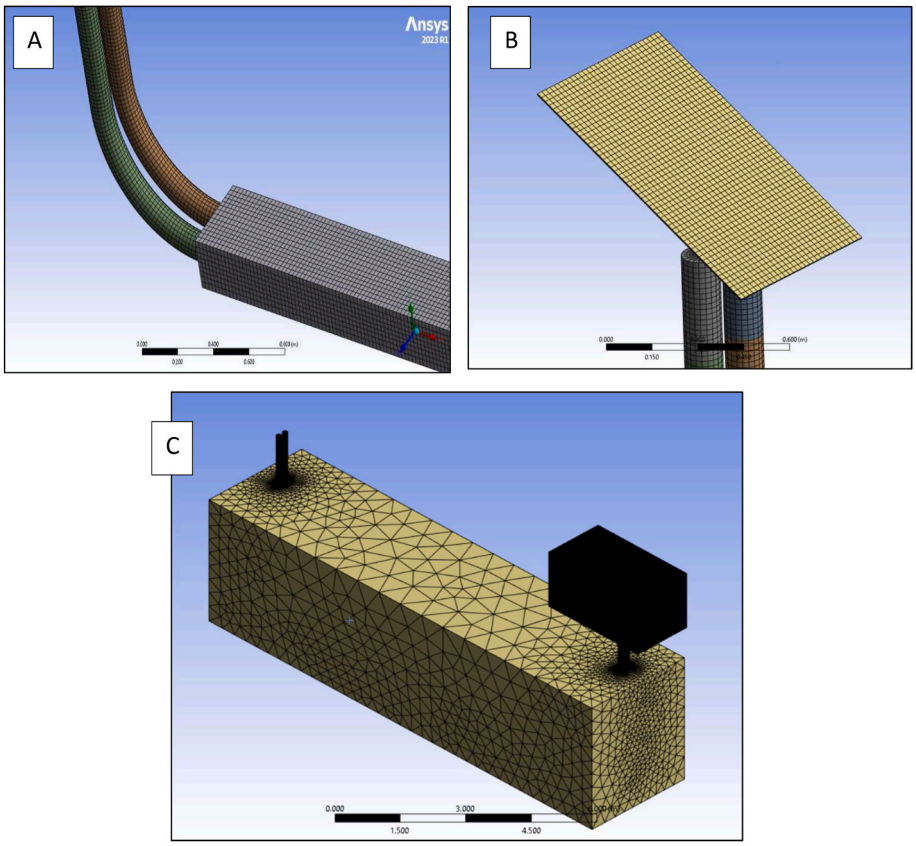


Fig. 2. Meshing of all the geometries: A. Piping and cement section, B. PV layers, C. complete design meshing of PV, piping, and ground.

Table 2
Boundary conditions for the system simulation

Location	Type	Value
Pipes inlet	Velocity inlet	$V_{in} = 1.5$ m/s, $T_{in} = 45$ °C
Pipes outlet	Pressure outlet	$P_{out} = 0$ gauge, $T_{out} = 45$ °C
Ground surfaces	Constant temperature	$T_{const} = 20$ °C
Pipes-bricks wall	Coupled-wall	–
Pipes-ground wall	Coupled-wall	–
PV outer surface	Solar radiation	From DO model

noon, while the lowest value, 360 W/m^2, was recorded at 5:00 p.m. Furthermore, Fig. 6 illustrates the air temperature gradient surrounding the system, with temperatures gradually rising until reaching a peak of 45 °C at 1:00 p.m. Both ambient temperature and solar radiation are integral to the simulation in ANSYS Fluent software. Incorporating these factors allows for the determination of the solar cell temperature, facilitating the calculation of solar panel efficiency and subsequent electrical power generation.

Figure (7) illustrates the PV temperature for both the PV-only and PV-ground cases, demonstrating a notable improvement in PV performance with the implementation of the cooling system. Specifically, the temperature decreased from 53.8 °C to approximately 42.8 °C at 1:00 p.m. This represents a significant enhancement, with the temperature reduced by approximately 28 % compared to the PV-only case.

Figure (8) displays the efficiency of the PV panel, revealing a decrease as the temperature rises. However, the PV-ground case consistently outperforms the PV-only case. At 8:00 a.m., the maximum efficiency was recorded as 16.53 % for the PV-ground case and 16.14 % for the PV-only case. By 1:00 p.m., the efficiency decreased to 15.49 % and 14.54 % for the PV-ground and PV-only cases, respectively.

Fig. 9 depicts the output power from the PV, indicating that the PV-ground configuration enhances power generation by approximately 6.5

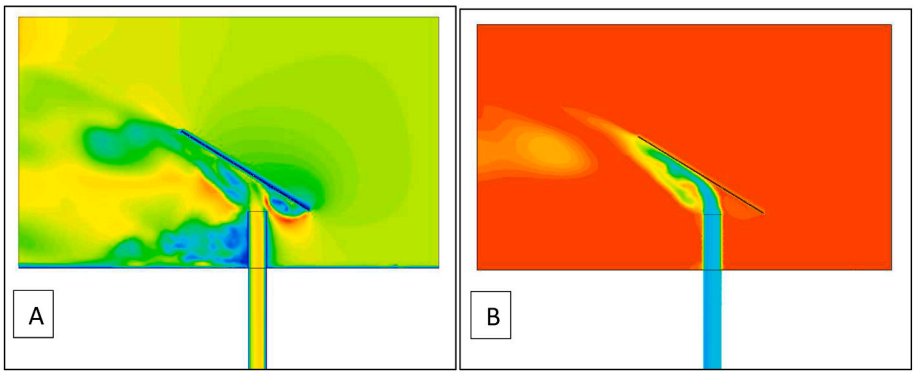


Fig. 3. A-Velocity and B- temperature, contour near the PV area.

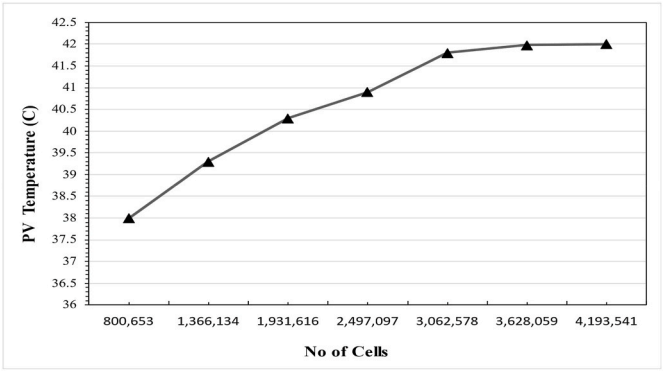


Fig. 4. Grid independence test (GIT).

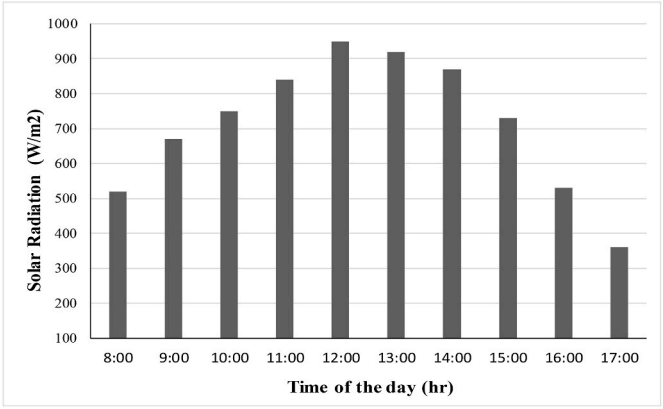


Fig. 5. Solar radiation values of the study location.

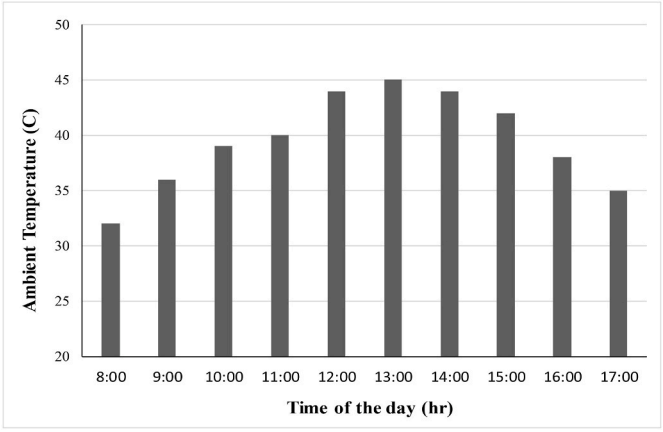


Fig. 6. Ambient air temperature of the study location.

% compared to the PV-only setup. The maximum power output achieved is approximately 88 W.

4.1. 4-1 comparison of photovoltaic (PV) temperature across studies

In this section, we analyze the temperature data collected from photovoltaic (PV) panels across various studies, with the corresponding illustration provided in Fig. 10. The study by Lopez-Pascual et al. [33] utilized a cooling method employing low-enthalpy geothermal cooling, utilizing a natural underground heat sink along with a water-based coolant to mitigate PV temperatures. They reported temperatures ranging from 31 °C to 48 °C for their investigation. The recorded

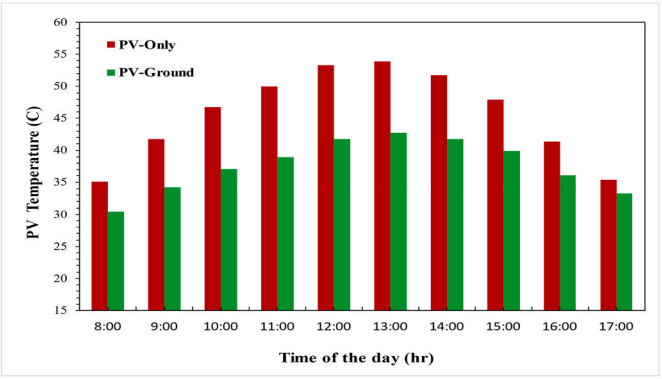


Fig. 7. Hourly PV surface temperature.

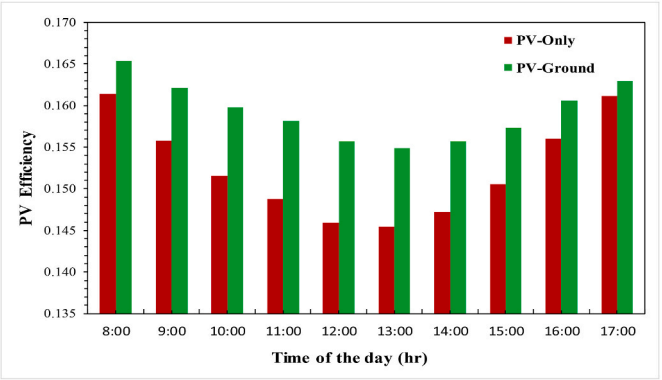


Fig. 8. Hourly PV efficiency.

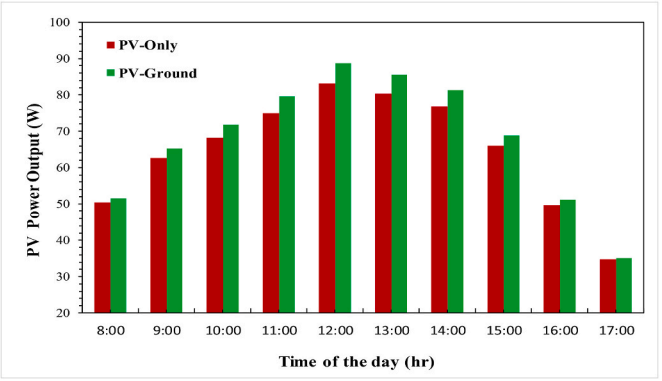


Fig. 9. Hourly output power of the PV.

temperatures suggest a moderate level of thermal management, with peak temperatures reaching 48 °C. In the study by Majeed et al. [34] who utilized an underground heat exchanger buried 4 m below the surface with a water-based coolant to achieve the cooling, PV temperatures exhibited a broader range, spanning from 33 °C to 57 °C and the highest recorded temperature in this study reached 57 °C, suggesting potential challenges in thermal regulation. In contrast, the present study introduced an innovative approach by harnessing underground sources of cooling air as the coolant. Our findings reveal a narrower range of PV temperatures, ranging from 32 °C to 45 °C. Compared to the previous studies, our results indicate more effective thermal management, with peak temperatures capped at 45 °C. This technique offers a distinct advantage by leveraging the natural cooling properties of underground air without the need for additional heat exchange mechanisms or

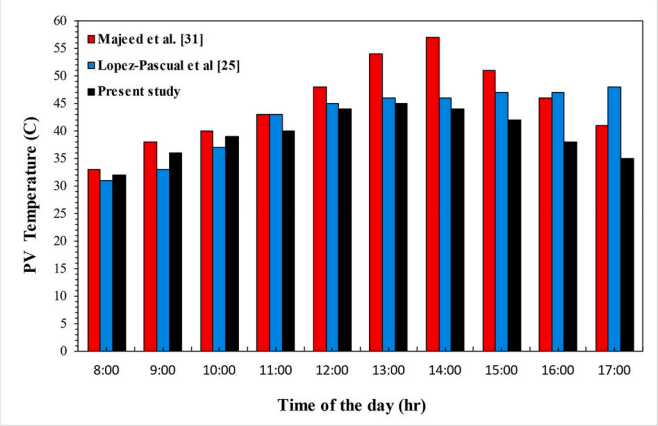


Fig. 10. Comparison of PV temperature.

extensive underground installations.

In addition to comparing the cooling techniques, it's essential to evaluate the efficiency of PV panels across different studies. Table 3 below presents the solar radiation, ambient temperature, and PV efficiency reported by various researchers, including the present study. Observing the PV efficiency across various studies reveals significant variations in performance influenced by solar radiation and ambient temperature. The present study demonstrates a noteworthy PV efficiency of 15.55 %, achieved under solar radiation of 950 W/m^2 and an ambient temperature of 44 °C. Comparatively, this efficiency surpasses several previous studies in the field. It's worth noting the impact of environmental factors on PV efficiency.

Studies such as Shalaby et al. [35] and Hasan et al. [36], conducted under higher solar radiation and ambient temperatures, achieved relatively higher efficiencies. Conversely, studies like Gomaa et al. [37], conducted under lower solar radiation and ambient temperatures, reported lower efficiencies. Overall, the efficiency achieved in the present study underscores the effectiveness of the cooling technique in optimizing PV performance under challenging environmental conditions. This reinforces the importance of innovative cooling strategies in enhancing PV efficiency and sustainability.

The contours for the velocity contour alongside the whole system are shown in Figure (11-a). Temperature contour alongside the whole system is shown in Figure (11-b) which shows that the air being sucked in from the ambient as hot air then as it goes along the pipes underground.

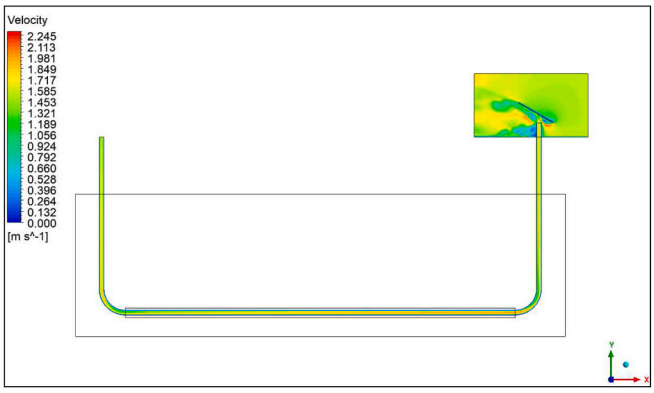
5. 5- Conclusion

Using Ansys Fluent 2023 R1 software and a detailed 3D model, we gained valuable insights into the thermal behavior of PV panels over 10 h, from 8:00 a.m. to 5:00 p.m. to study the PV performance with cooled air by ground source energy in Kirkuk city, northern Iraq (35° 28′ N - E. 44.39). The most significant observation from this study was the remarkable reduction in PV panel temperature achieved through the proposed cooling system, with a notable temperature improvement of approximately 28 % compared to conventional cooling methods. This enhancement in temperature control has the potential to significantly prolong the lifespan of PV panels and boost their overall performance. Furthermore, the comparative analysis of PV panel output power demonstrated a substantial improvement of approximately 6.5 % when employing the PV-ground configuration, ultimately reaching a maximum power output of approximately 88 W. This enhancement in power generation is a critical factor for improving the overall efficiency of PV systems, particularly in regions with high solar radiation. This cooling mechanism resulted in a significant reduction in air temperature, ultimately leading to a substantial decrease in PV panel temperatures.

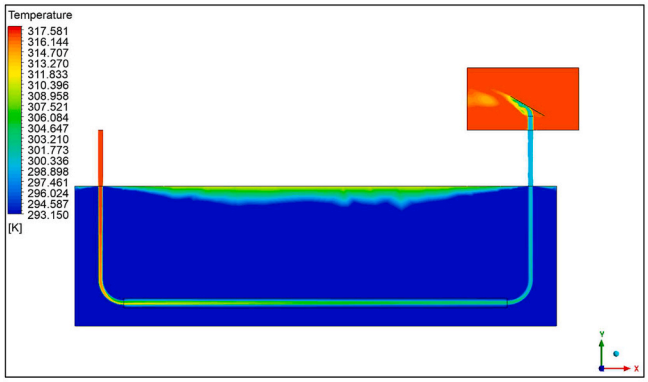
By comparing the cooling techniques along with the fluid used, it is evident that while all approaches aim to mitigate PV temperatures, the present study's method offers simplicity, cost-effectiveness, and environmental sustainability. Harnessing underground cooling air as the coolant presents a promising alternative to conventional cooling methods, offering potential benefits in terms of efficiency, durability, and scalability. This comparison underscores the innovative nature and potential superiority of the cooling technique employed in the present study, paving the way for further exploration and implementation in the field of PV technology.

Table 3
Efficiency comparison across various studies

References	Solar radiation (W/m^2)	Ambient temperature (°C)	PV Efficiency (%)
Dubey and Tiwari [38]	850	38	9.5
Shalaby et al. [35]	1087	43	13
Chow et al. [39]	771	32.1	11
Hasan et al. [36]	840	45	13.8
Menon et al. [40]	900	36	15
Majeed et al. [34]	833	49.22	11.82
Gomaa et al. [37]	1000	29	9
Ji et al. [41]	245.6	32.4	9.87
Salem et al. [42]	800	35	13
Kazem et al. [43]	773	37	10.8
Present study	950	45	15.55



(a) Velocity contour alongside the whole system



(b) Temperature contour alongside the whole system

Fig. 11. Temperature distribution and velocity profile for the system.

CRediT authorship contribution statement

Emad Jaleel Mahdi: Investigation. **Sameer Algburi:** Supervision. **Nasser Al-Abadi:** Methodology. **Omer K. Ahmed:** Data curation. **Amer K. Ahmed:** Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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